MOUNT MORGAN IS F*#KED

Mining is the lifeblood of our beautiful country, but if left unchecked, it will also be the poison with which we'll choke ourselves.

Sam Brown Jacob Hatwell Thomas Bateman Kaijun Liang
Brodie Robertson Dineth Hirun

Mount Morgan Mine

Originally started in 1882, the mining town of Mount Morgan was one of the most prosperous in the country, up until it's shutdown in 1990 (Calderwood, 2014).

However, this shutdown and rehabilitation was never completed, and as a result, contaminates from the mine have been leaking into the local ecosystem, under the watchful eye of the Queensland Government (Queensland Goverment, 2021).

Over 100 years of uninterrupted mining, while commendable for its economic prosperity, it's doubtlessly going to result in a large amount waste product.

Even as early as 2006 there was great concern about the groundwater contamination (Christoph Wels, 2006). These fears were well grounded, and unfortunately proved to be only the start. In 2013, the Flooded Open Pit Mine overflowed, spilling tons of contaminated across the landscape and into the river.

"the Dee River is an unnatural shade of blue-green for a lot of its length, and birds and fish are dying." (Townsend, 2013)

The Evidence

"The fish all died. It smelled. We obviously knew the water was toxic" (Terzon, 2018)

"It was pale green, nearly luminescent" (Terzon, 2018)

Studies conducted on the effects of the site show that the local Dee river contains contamination levels hundreds of times stronger than what can normally be expected in the region, which has resulted in the untold death of thousands of aquatic species. (Edraki, Lawrence, Baublys, & Golding, 2005)

Sample number	Location	pH	SO_4	Al	Ca	Cu		Co	Fe	Mg	Mn
Dee River											
21	Carmody bridge	7.0	340	BDL	125	0.00	BDL	BDL	63	0.24	2.49
15	No. 6 A Dam	6.8	410	BDL	98	0.02	BDL	BDL	58	0.06	4.22
14	No. 6 Dam	3.2	3910	147	169	9.08	0.42	17	522	22.53	8.09
12	No. 4 Dam	3.1	4750	169	231	5.66	0.47	23	651	29.77	10.63
13	No. 5 Dam	3.1	3800	76	183	2.45	0.20	17	561	23.51	10.09
17	Tipperary Point	3.6	2200	50	161	2.15	0.16	3	311	18.88	7.99
18	Red Hill Crossing	2.7	5950	390	309	45.03	0.99	74	693	45.87	5.08
19	Curtis Hole	2.9	5880	347	343	35.67	0.93	28	715	46.96	6.06
20	Piebald Crossing	6.2	340	1	66	0.06	BDL	BDL	74	0.58	3.22

(Edraki, Lawrence, Baublys, & Golding, 2005)

- This Table shows that...

the level of chemicals are up to 2.5 times greater than normal and resembles more of a toxic waste then a thriving river system.

WHAT WILL HAPPEN TO THE **water?**

From more than a century of gold and copper mining, over **100 million tonnes** of waste material including sulphide, heavy metals and acids has been spread throughout Mount Morgan (Gasparon et al. 2007). These pollutants gradually seep into nearby waterways through a mechanism called **acid mine drainage**. This occurs when water comes into contact with those toxins, causing the water to become acidic and release said pollutants into the larger bodies of water such as ponds, rivers and lakes.

In addition to acid mine drainage, the ongoing seepage of **tailings from the mine site** can also lead to the accumulation of hazardous substances in nearby waterways. The tailings, which contain heavy metals and other pollutants, can gradually seep into groundwater and nearby streams and rivers, leading to greater contamination of the water over time (Goldsmith, 2014).

Moreover, the effects of climate change can worsen this problem by interfering with the natural flow and quality of water in the region. Droughts can cause decreased waterflow, causing heavy metals and other dense pollutants to build up in smaller volume of water. Heavy rains and floods can also worsen the problem by naturally releasing more pollutants into waterways. For example, the floods of 2011 caused a massive amount of toxins to gather in the nearby **Dee river**, which turned its color into a deep blue (Townsend, 2014).

AS FOR

human health,

When heavy metals pollute the water, they have entered the human food chain through aquatic life. Many ailments, such as diabetes, Alzheimer's disease, and other cancers, are on the rise, and heavy metals are directly responsible for this (Jara-Marini, Molina-García, Martínez-Durazo, & Páez-Osuna, 2020). In order to demonstrate the harm of heavy metals in more detail, Arsenic and chromium - which are some of the main elements in the polluted water are mentioned below (Wang, Gao, Jiang, Zhang, & Liu, 2021)

Arsenic
Arsenic is produced during the smelting of metals such as copper and zinc. Long-term ingestion of arsenic will damage muscles and blood clotting proteins which have negative effects on the nervous system. Moreover, excessive arsenic affects the endocrine system, liver system, and reproductive system (Sankhla, Kumari, Sharma, Kushwah, & Kumar, 2018).

Chromium
The metal is destroying natural water resources because of mining processes. According to research, Long term intake of Chromium will harm the recycling system and nerve tissue (Hossini, et al., 2022). In addition, overmuch Chromium will increase the load of the lung which may lead to lung cancer (Ishikawa, et al., 1994).
(Bada, 2019)

WHAT IS BEING

DONE?

Land-based and floating evaporators to lower water level in the mine...

Sealing Dam 8 upstream wall...

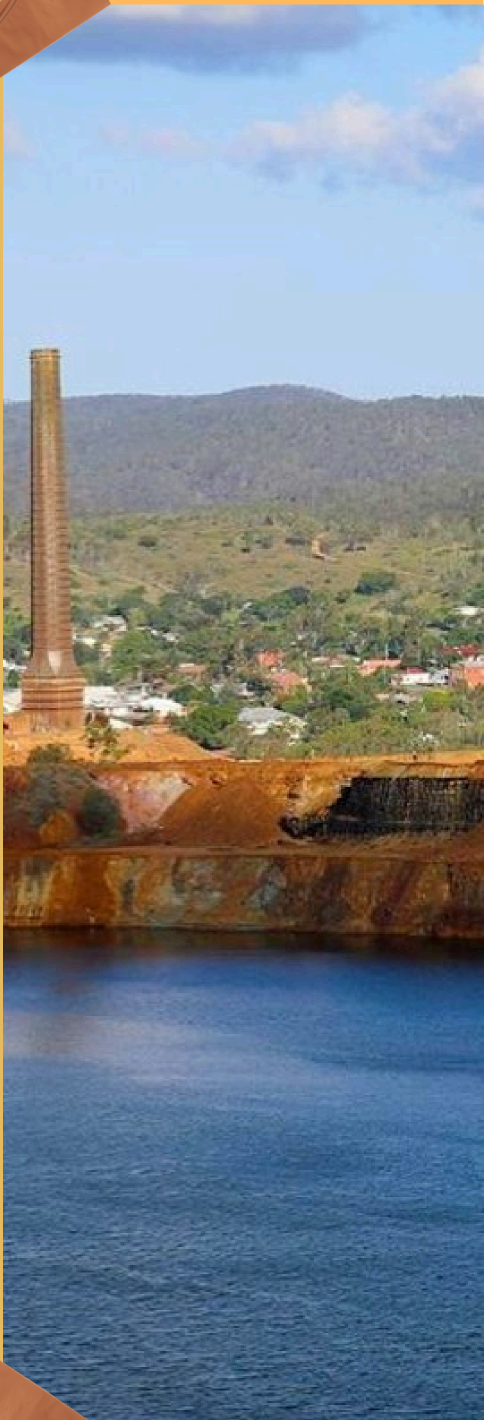
Routinely operated stream and ground water testing...

Water contamination notification system for downstream landowners...

Upgraded high voltage electrical lines to aid future work...

This, is what the Queensland government has begun to do in efforts to rehabilitate Mount Morgan to its former glory. **But is it enough?**

Partnering with Heritage Minerals has allowed the local government to begin to preserve historical elements of the mine. Historical significance brings many tourism opportunities to the area. However, tours to the caverns, some containing fossilised dinosaur footprints, were stopped in the past 15 years due to safety concerns.



CALL TO ACTION

Mt Morgan, a mountain destroyed for the Governments benefit left an ugly hole in the Queensland terrain. Being left unfilled, this lake of now contaminated rainwater has been dwelling in our lands for far too long. The government has proven to be doing very little to solve this significant environmental issue thus creating dire health concerns. Each year rain continues to lead to an overflow of this contamination leaving nearby land and natural habitat plagued with mercury and cyanide. Something has to be done to bring back Australia's pristine environment!